CS 255 Model Application Short Paper

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Process Model Application

First, for process modeling, we need to define what it is. In summary, it is a model that displays the flow of the system. At the same time, it explains their processes/functions that can have different conditions. One diagram used in process models is activity diagrams. In activity diagrams, they display the flow of the system with lines, conditions, and activities (from where each line connects). Because there are a lot of scenarios for each user, there have to be many paths displayed.

Within the Driver-Pass transcript, there is a lot of mention about how courses are managed and appointed. For example, the owner Sam stated how customers can make reservations or appointments. If any customer wants to make a reservation online (using their account) or call the office, they will need to share information to be updated in the system. After that, the location of the pickup and drop-off of the customer will be available for them to choose from. Then, the system will have to identify the driver for the customer scheduled to go with. If the customer forgets their login information, they can automatically reset it.

All of what was explained can be shown in an activity diagram; at the top it will have the initial steps towards using the web. Customers will login or register (depending on the situation), then it will take them to plan out what they want out of Driver-Pass. Once all the information has been processed and courses begin, the interface can display the course's progress, notes, special needs, and information of the customer. This diagram gives a clear understanding of how the website should be that facilitates developers and the owner, Sam's vision.

Object Model Application

Object modeling is different from process modeling. In object modeling, the goal is to create a system design that reflects objects inside a structure framework that contains their attributes and

relationships towards each other. For example, some objects can be stable and their ties with other objects are unchangeable. On the other hand, the system can change their state and their interactions over time. In Driver-Pass, there are a lot of objects and behaviors that are relied on each other.

Liam stated how the interface would have elements from the courses, like a status. He explained, "As you can see, the online test progress should show the tests the customer took". To be able to see the progress of the online test, there would be an online test and a course to host that test. This is an established dependency between the objects online test progress bar and the online test itself. His vision of the interface can be displayed by using a UML class diagram, it displays the objects within the class, how their relationships work and what is contained in them which can be final (unchangeable), static (the object shared by all instances of the class), or changeable.

Process and Object Model Comparison

Both Models have their specific purposes and what they want to convey. As mentioned before, process modeling is the model that defines the function of the system and the blueprint of how all the mechanic's work. Using an activity diagram details the flow of the system which gives everyone that includes developers, stakeholders, and the owners, a clear idea of how everything will be laid out, this can be seen like how Liam already had a layout planned so having a diagram will make it a lot easier for him to be onboard. When the design is fully understood, everyone can come to a clear agreement while being fully onboard. On the flip side, process models can be either too rigid or vague in their designs, which means they can always change drastically and make the model divert the focus on behaviors, not the structure.

Object modeling, however, simplifies the implementation of objects and their relationships by reducing the ambiguity of the layout. Objects like methods are defined (like UML class

diagrams); they can also be reusable in different objects facilitating maintenance as well. In other words, it is more in depth but simple for everyone. However, maintainability means that the system requires that maintainability to be consistent as the system evolves or changes. So, maintenance can vary a lot to where it is necessary to modify classes, objects and relationships, changing the diagram. There is also a big learning curve like how the UML class diagram uses object-oriented programming (OOP) to understand the relationships between objects and classes.